

Shallom High School

"Wisdom, Peace, Excellence" — Java Swing Desktop Portal

README

1. What this project is

A role-based desktop application for Shallom High School, built with plain Java and Swing — no external frameworks, no build tool, and no database. One login window authenticates six account types — Headmaster, Deputy Headmaster, Teacher in Charge, Teacher, Ancillary Staff and Student — and opens a dashboard window whose navigation and content change depending on who signed in. All school data (subjects, facilities, timetable, class teachers, fee balance, login accounts) is stored in memory as static Java fields.

2. Technology stack

- **Java (JDK 17 or later)** — no other language runtime required
- **Swing** (javax.swing, part of the standard JDK) for every window and widget
- No Maven, Gradle, or third-party libraries — the project compiles with javac alone
- No database — data lives in static fields inside SchoolData.java for the lifetime of the run

3. Project files

All four files sit directly in src, with no package declarations and no sub-folders:

Main.java — entry point; opens the login window

LoginFrame.java — login window: role dropdown, User ID, password, validation

DashboardFrame.java — post-login window: role-based side navigation and content panels

SchoolData.java — static data: accounts, subjects, facilities, timetable, class teachers, fees

4. Running the project in IntelliJ IDEA

- Create or open a plain Java project (no build system needed).
- Copy all four .java files directly into the src folder.
- Open Main.java and click the green Run arrow next to the class or the main method.
- The login window opens immediately — there is no server to wait for and no browser involved.

5. Demo login credentials

Role	User ID	Password
Headmaster	hm001	head2026
Deputy Headmaster	dh001	deputy2026
Teacher in Charge	tic001	tic2026
Teacher	tch001	teach2026
Ancillary Staff	anc001	staff2026

Student	stu2026001	student2026
---------	------------	-------------

Each account only sees the navigation options and panels relevant to its role — for example, the Teacher in Charge account sees the Form 4A class register, while the Student account sees a personal fee balance and payment methods.

6. What each role can see

Role	Dashboard sections
Headmaster	Overview, Master Timetable, Staff & Class Teachers, Subjects, Facilities, Fees & Payment
Deputy Headmaster	Overview, Master Timetable, Staff & Class Teachers, Discipline Log, Facilities
Teacher in Charge	My Class, Class Timetable, Class Register, Subjects Offered
Teacher	Overview, My Timetable, My Classes, Subjects Offered
Ancillary Staff	My Duty, Duty Roster, Facilities
Student	Overview, My Timetable, Subjects Offered, School Fees, Facilities

7. Notes for markers / future work

- Login accounts are hardcoded in SchoolData.java as a `Map<String, String[]>`; a production version would replace this with a database table and hashed passwords.
- There is no persistence — closing the app discards any in-session state, since none of the panels currently write back to SchoolData.
- The timetable shown is a single sample class (Form 4A); the Class Teacher Assignments panel covers all ten classes.
- Extending this to JavaFX or a web front end would mean replacing LoginFrame/DashboardFrame while keeping SchoolData largely as-is.